HCDE 511

Information Visualization

Winter 2026

4

Week

Week 4

Project proposal pitch + Q&A (1.5 hr)
Break - 20 min
User Perception Pt. 1 (1 hr)
Break – 5 min
Group work time (45 min)

Week 1	2	3	4	5	6	7	8	9	10	Finals
Intro Data encoding & mapping	Graphical excellence & integrity Critiquing	Fundamen-tal graphs Working with data	Lab: Tableau & Dashboard Visual perception Group time	Visual perception, cont. Tasks & inter-actions	Collabora-tive critique	Multi-variate analysis	Trees & networks	Animation, context, story-telling	Final pre-sentations	

A1 Good/bad

A2 Static vis

A3 Exploratory analysis

Project proposal

Rev. prop.

Final project deliverable

Final project writeup

Project Proposal Critique

2-3 Slides

1. Outcomes statement
2. What datasets are you looking at?
3. An early sketch of what you might make

4 min Pitch

1. Why did you choose this topic?
2. What are you planning to make?
3. Based on what your users told you, how are you going to design it?

2-3 min Q&A

1. Unanswered questions
2. Biggest challenges
3. Relevance to the users / world

Outcomes Statement

The purpose of ______

is to ______ so that ______

Pitch

Get feedback

Incorporate feedback in revised proposal (due next week)

P2

Revise your proposal based on feedback from today.

Explain why you revised it.

Break

until 7:30p

Visual Perception
Part I

Thinking with Your Eyes

Biology of Vision

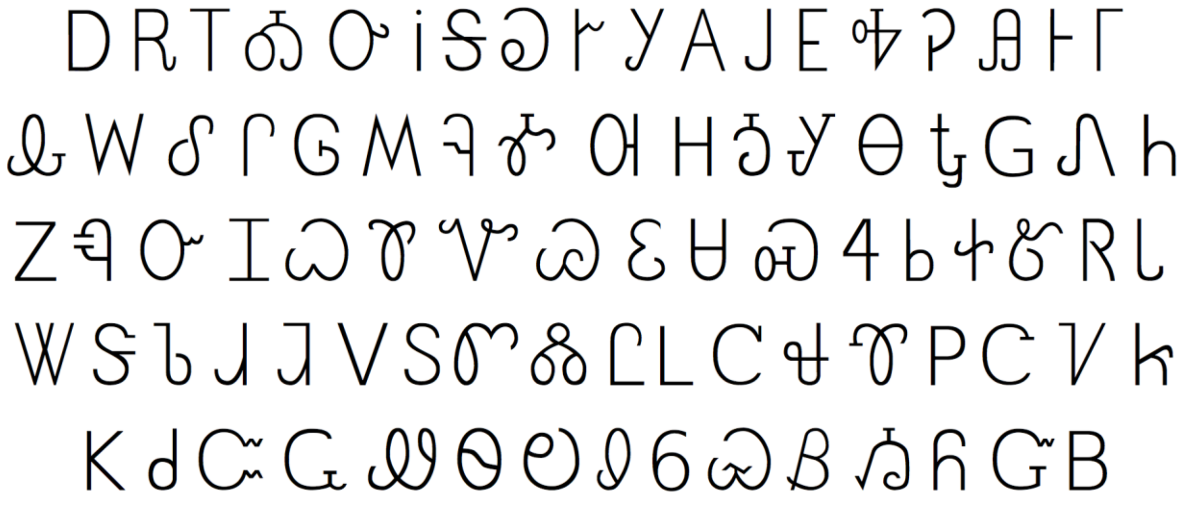
Preattentive vs. Attentive

Detection

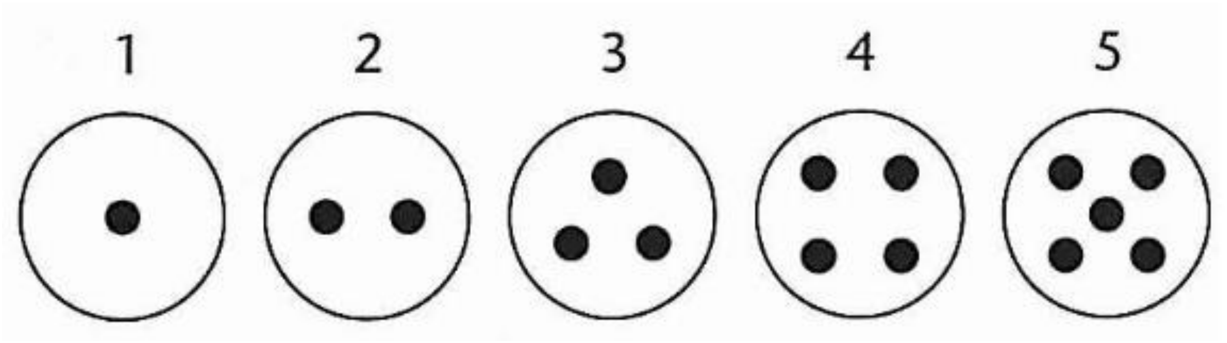
Arbitrary Representations

Hard to learn
Easy to forget
Embedded in culture and applications
Formally powerful
Capable of rapid change (versus evolution)

The Cherokee Alphabet



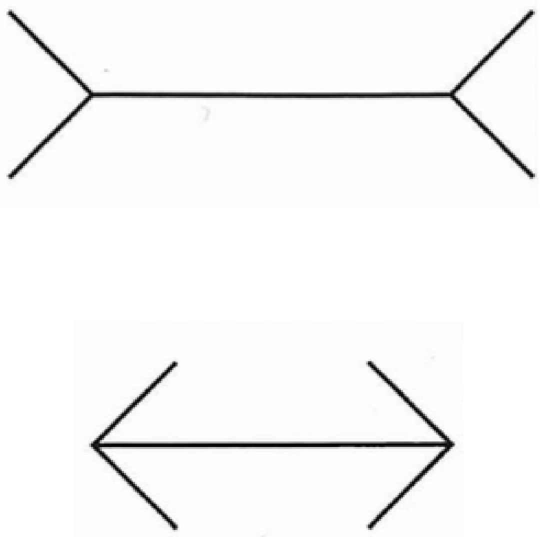
Two methods for representing the first five digits.

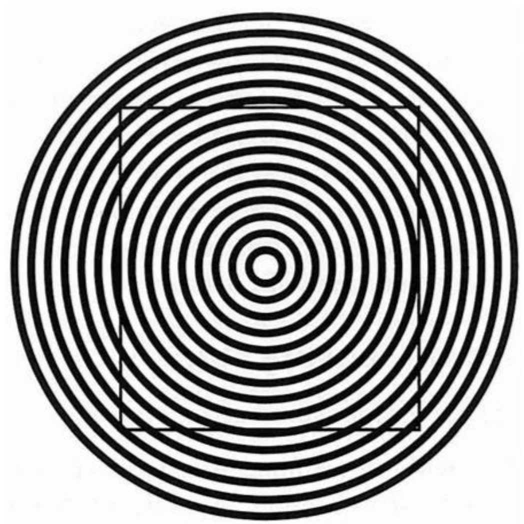


Sensory Representations

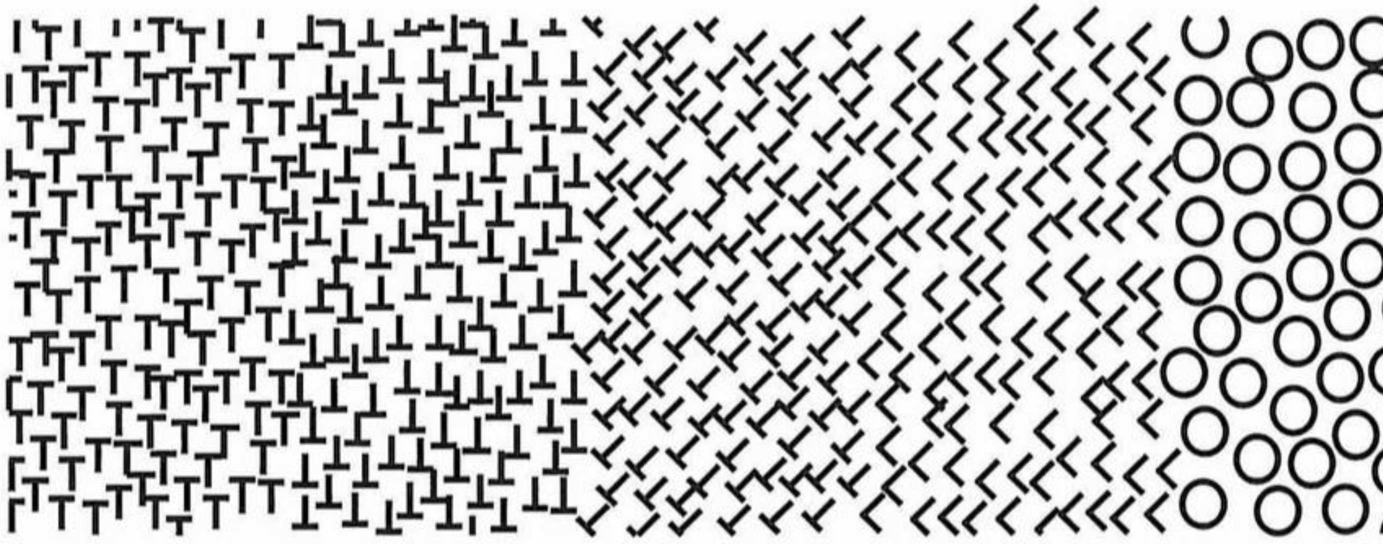
Understanding without training
Resistance to instructional bias
Sensory Immediacy
Cross-cultural validity

The Muller-Lyer Illusion





Five regions of texture. Some are easier to differentiate visually than others. Adapted from Beck (1966).



Why Visual perception matters

Exploit strengths, avoid weaknesses

Optimize, not interfere

Inform Design criteria

Expressiveness

Effectiveness

No false messages

Thinking with our eyes

70% of the body's sensory receptors reside in our eyes

Metaphors to describe understanding often refer to vision
("I see," "insight," "illumination")

"The eye and the visual cortex of the brain form a massively parallel processor that provides the highest-bandwidth channel into human cognitive centers."

Colin Ware, Information Visualization, 2004

Thinking with our eyes

Working memory is extremely limited.

How might we overcome this?

"The process of grouping simple concepts into more complex ones is called chunking."
Ware, 2004

"The process of becoming an expert is largely one of learning to create effective chunks."
Ware, 2004

“It is possible to have a far more complex concept structure represented externally in a visual display than can be held in visual and verbal working memories.”

Colin Ware

Thinking with Your Eyes

Biology of Vision

Preattentive vs. Attentive

Detection

How the eye works

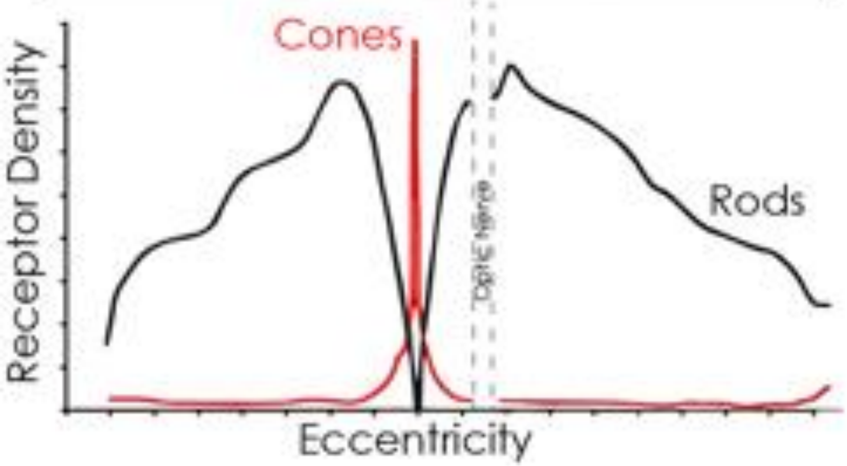
The eye is not a camera!

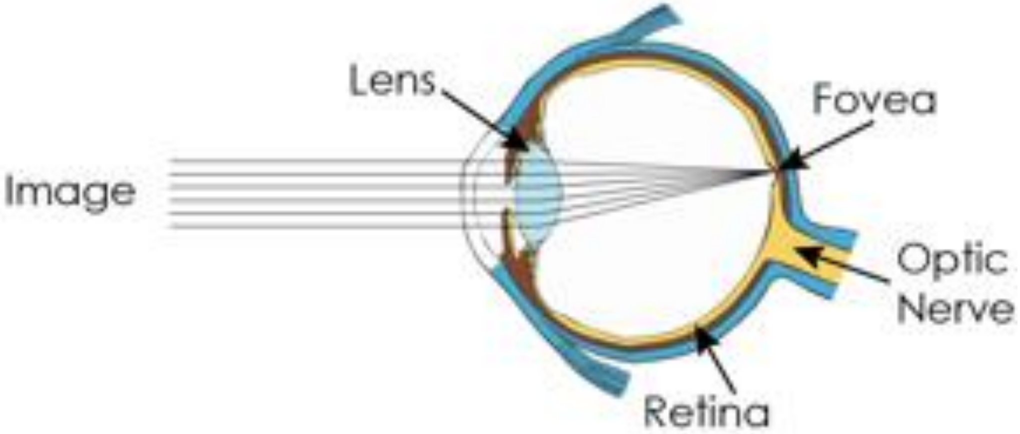
Better metaphor for vision: "dynamic and ongoing construction project" – Healey

Attention is selective (filtering)

Cognitive processes modify

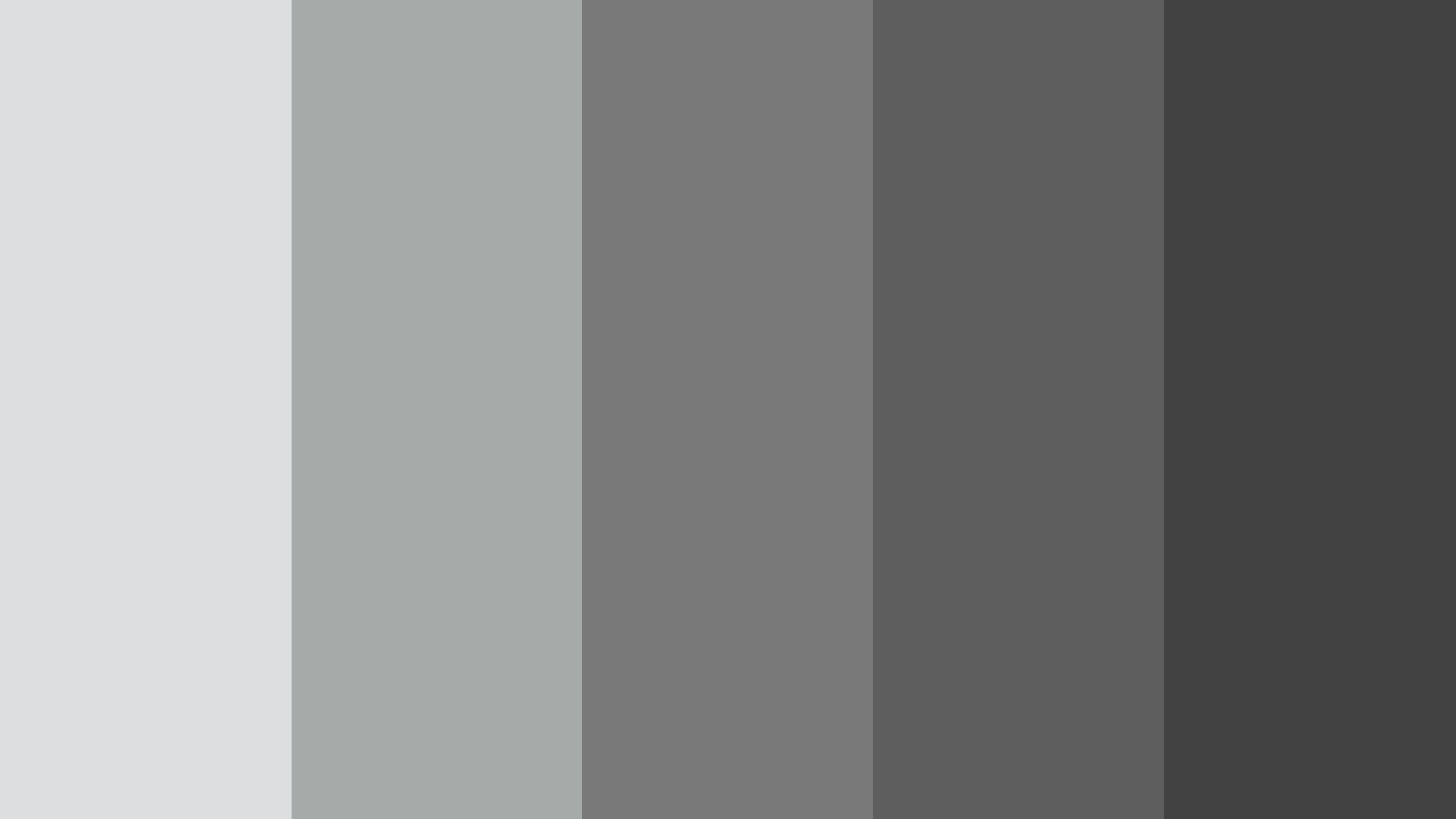
Psychophysics: concerned with establishing quantitative relations between physical stimulation and perceptual events

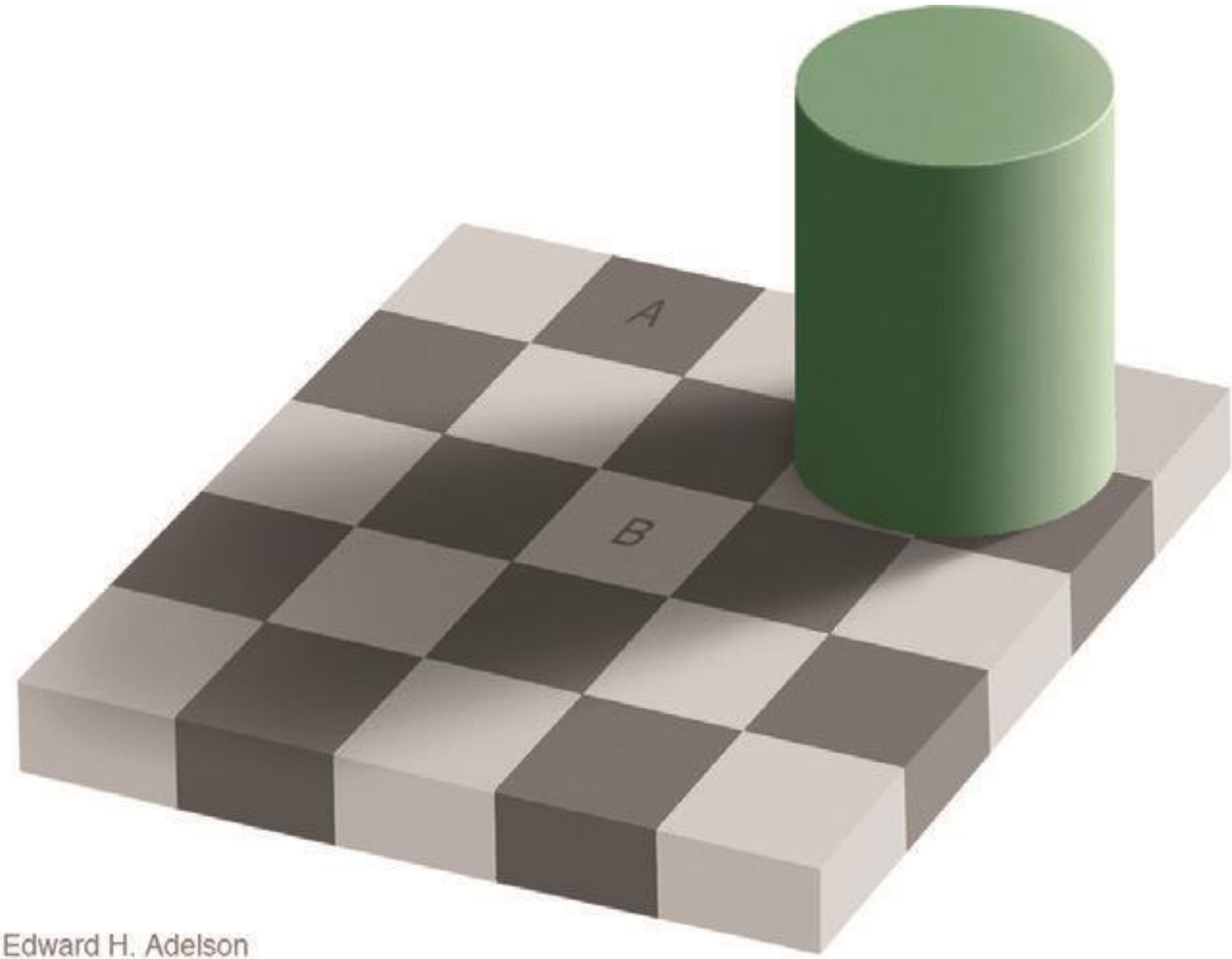

Cones
Receptor Density
Optic Nerve
Rods
Eccentricity


Lens
Fovea
Image
Optic
Nerve
Retina

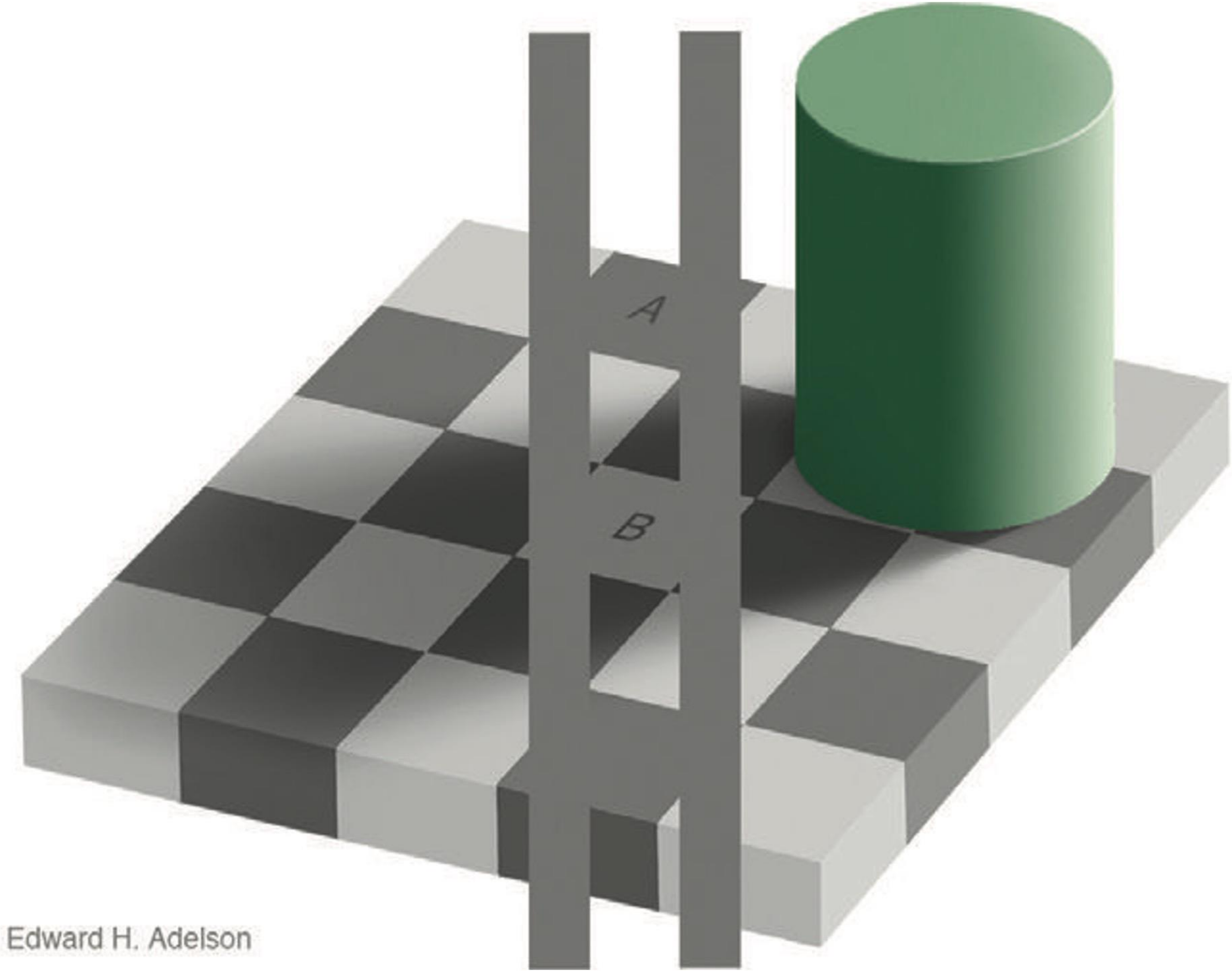
How we perceive a scene

Fixations and Saccades (~20-100ms)
Foveal focus shifts up to 25 times per second





Edward H. Adelson


A
B
Edward H. Adelson

Eyes

Relatively poor optics

Constantly scanning (saccades)

Constantly adjusting focus

Constantly adapting (white balance, exposure)

Mental reconstruction of image (sort of)

vs.

Cameras

Good optics

Single focus, white balance, exposure

“Full image capture”

Aoccdrnig to a rscheearch at Cmabrigde Uinervtisy, it deosn't mttaer in waht oredr the ltteers in a wrod are, the olny iprmoetnt tihng is taht the frist and lsat ltteer be at the rghit pclae. The rset can be a toatl mses and you can sitll raed it wouthit porbelm. Tihs is bcuseae the huamn mnid deos not raed ervey lteter by istlef, but the wrod as a wlohe.

https://www.mrc-cbu.cam.ac.uk/people/matt.davis/cmabridge/

RED

GREEN

BLUE

PURPLE

ORANGE

PURPLE

ORANGE

GREEN

BLUE

RED

Thinking with Your Eyes
Biology of Vision
Preattentive vs. Attentive
Detection

How many 5s?

385720939823728196837293827

382912358383492730122894839

909020102032893759273091428

938309762965817431869241024

How many 5s?

385720939823728196837293827

382912358383492730122894839

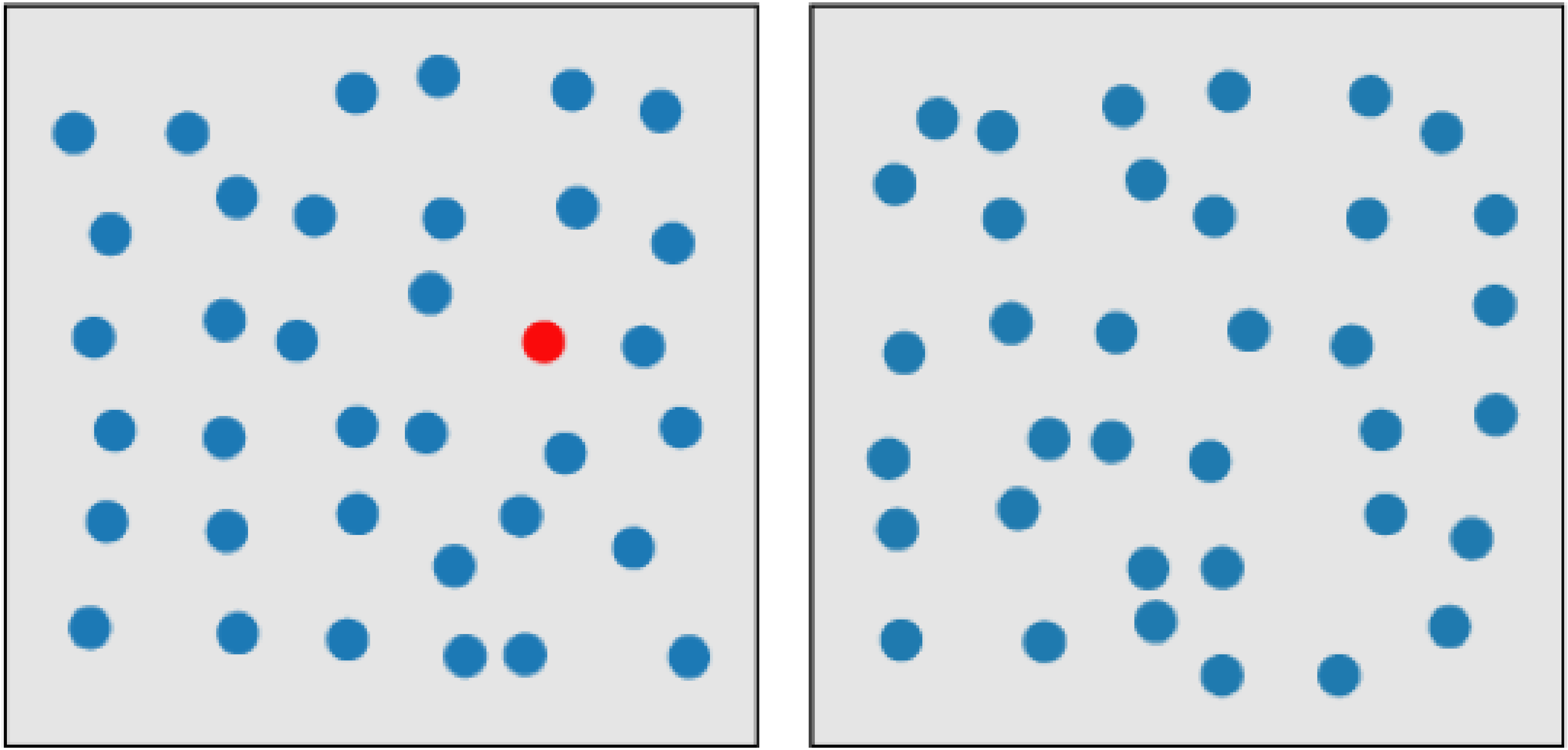
909020102032893759273091428

938309762965817431869241024

A limited set of visual properties are processed pre-attentively.

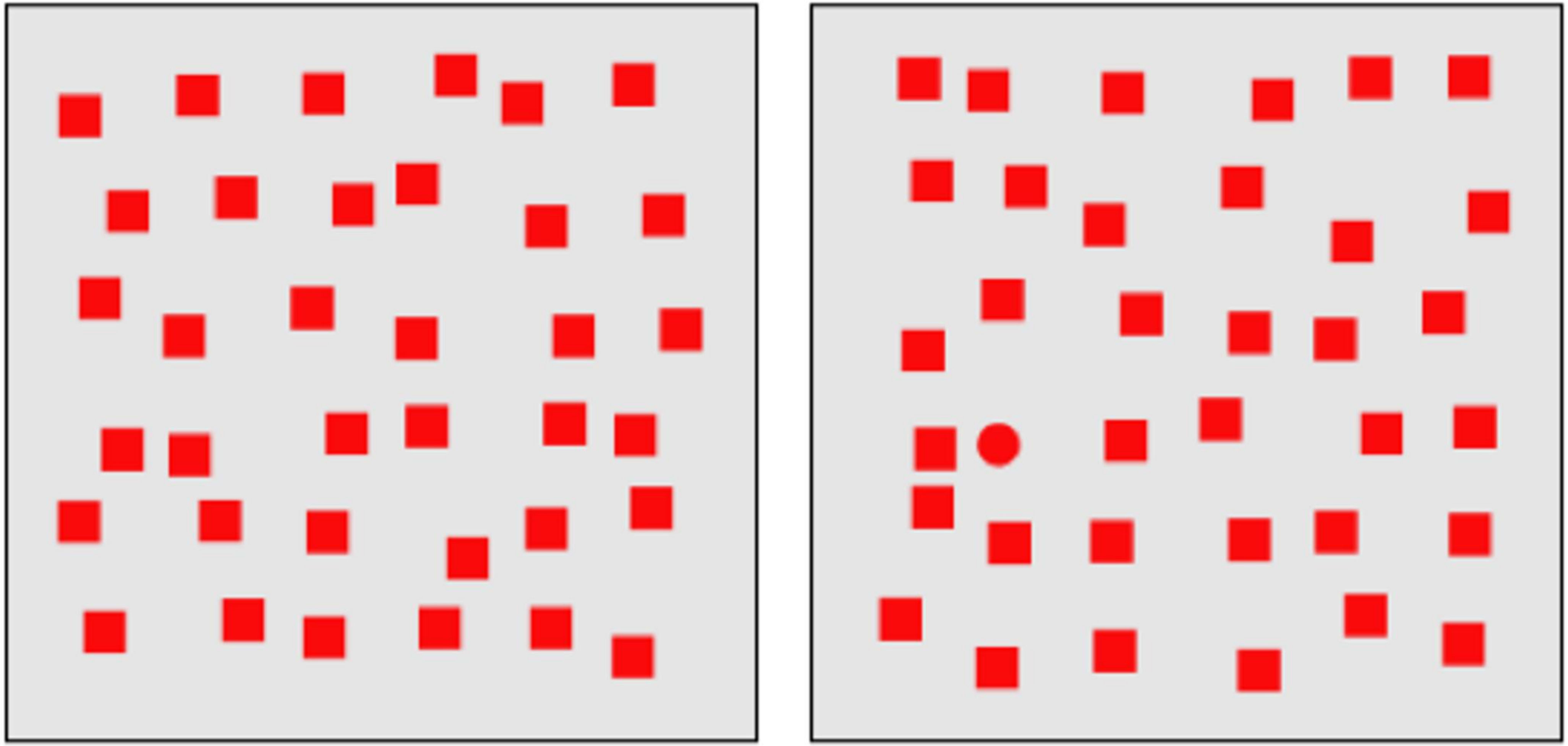
This is important for the design of visualizations.

Color



Healey 2005

Shape

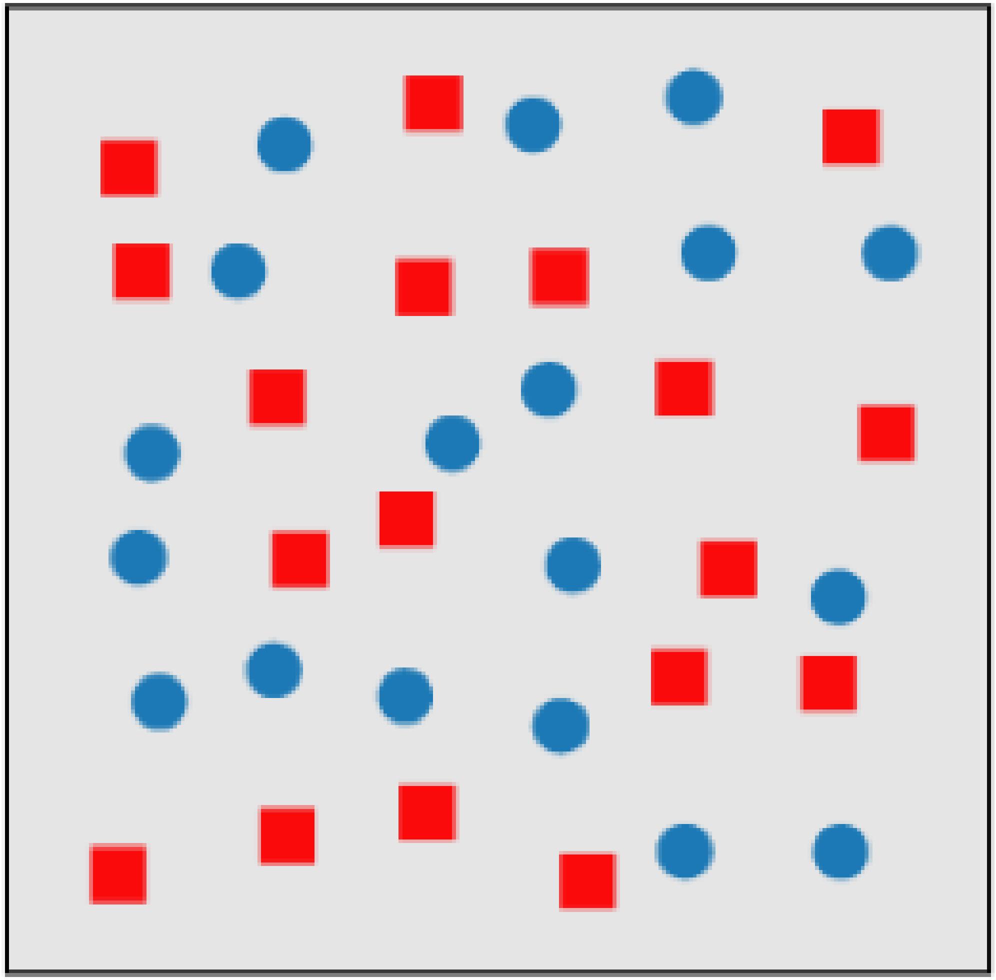


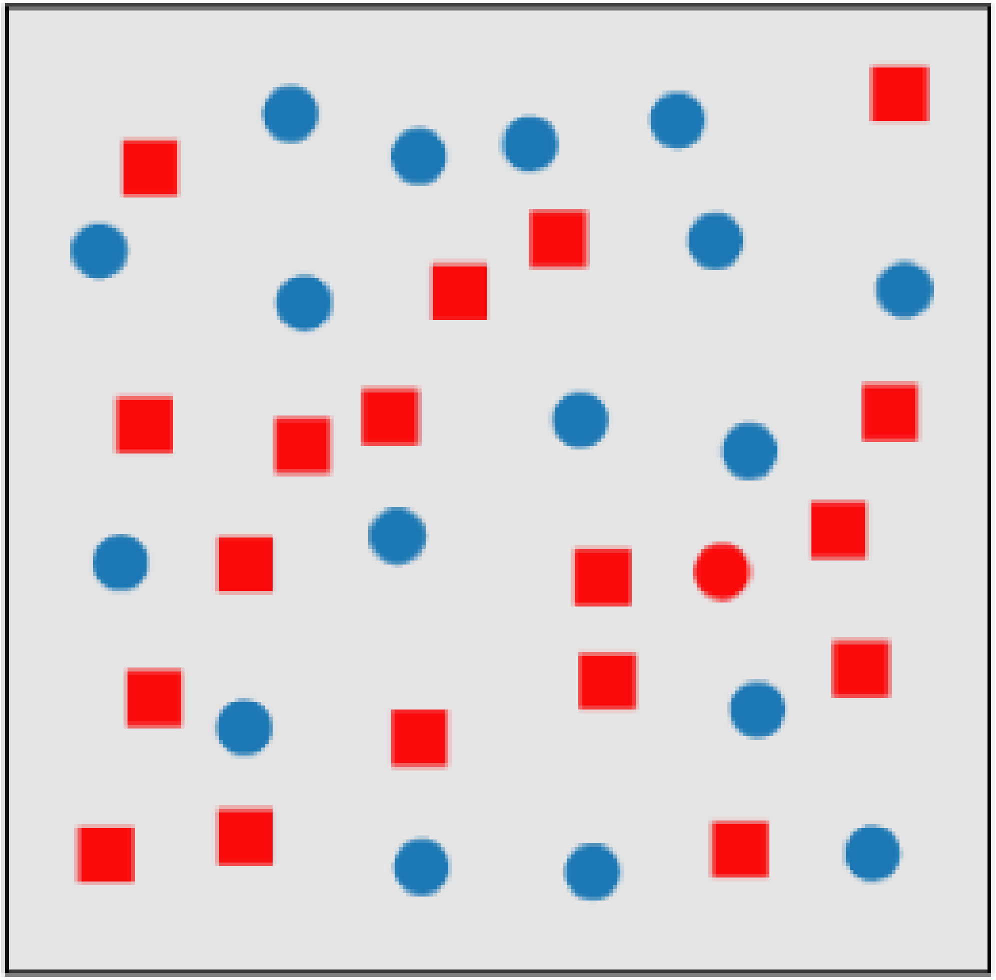
Healey 2005

< 200 - 250ms qualifies as pre-attentive

If a decision takes a fixed amount of time regardless of the number of distractors, it is considered to be preattentive.

Conjunction of Attributes



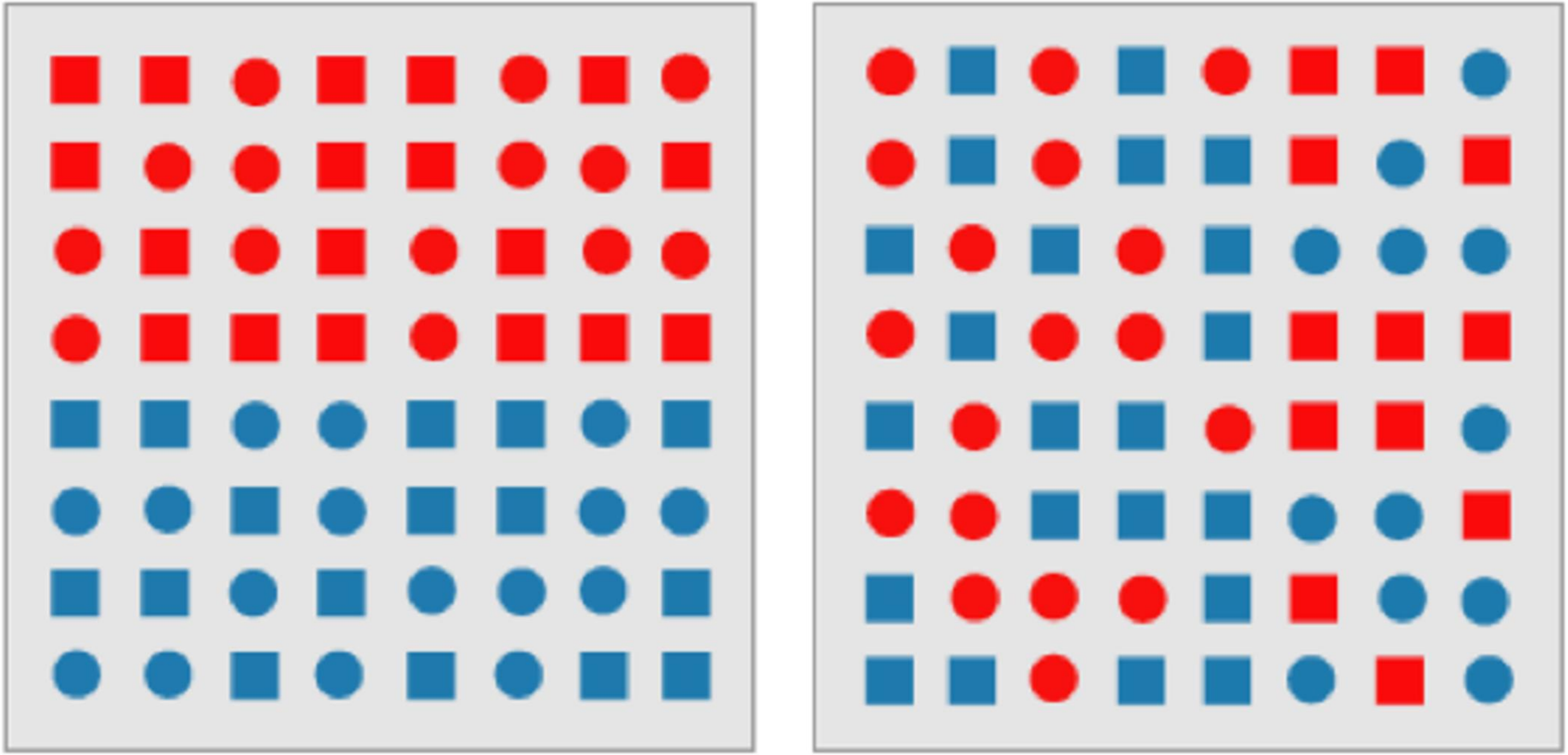


Healey 2005

Before

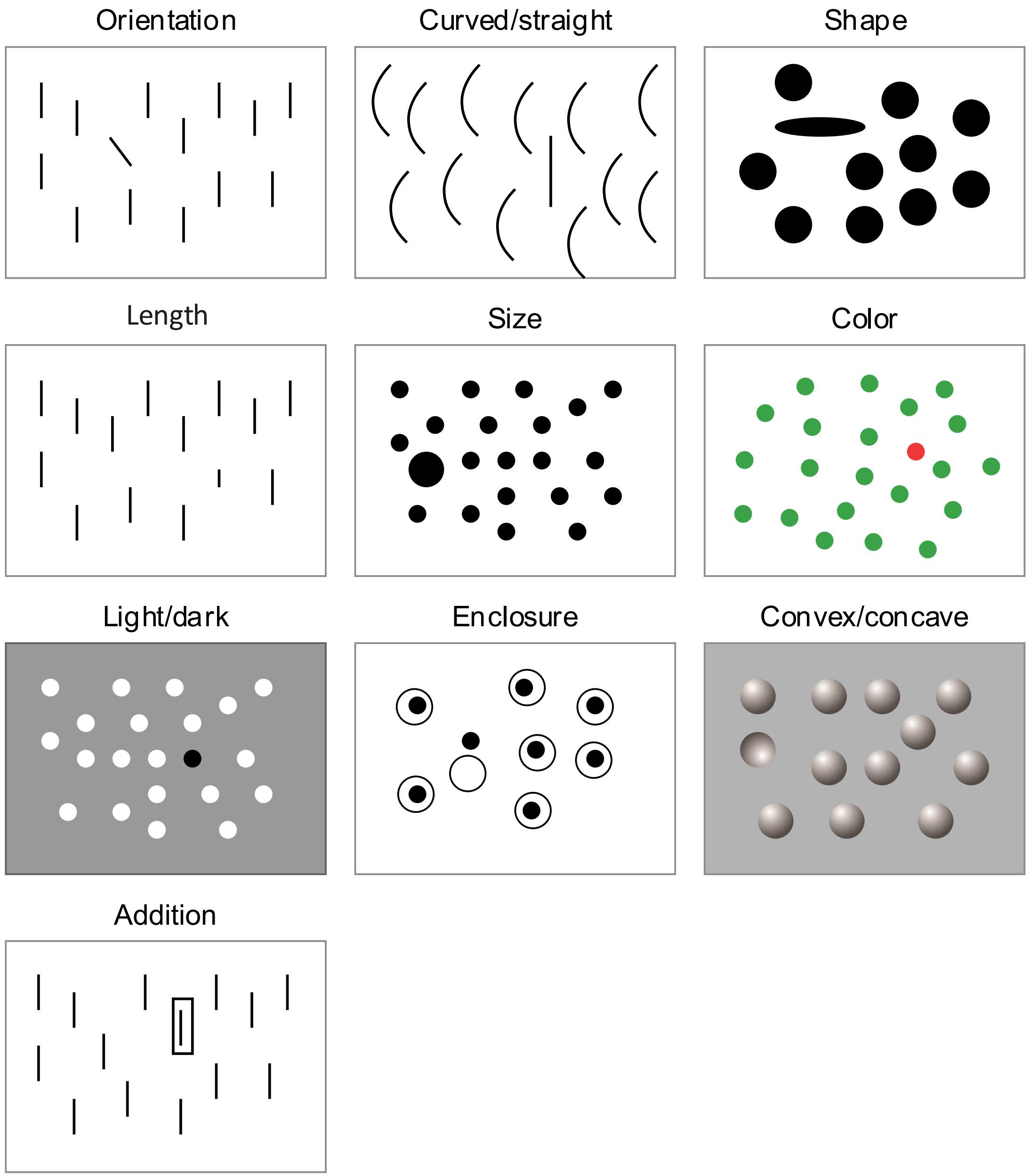
After

Conjunction of Attributes



Healey 2005

Preattentive Visual Features



Information Visualization 3rd edition. Figure 5.1

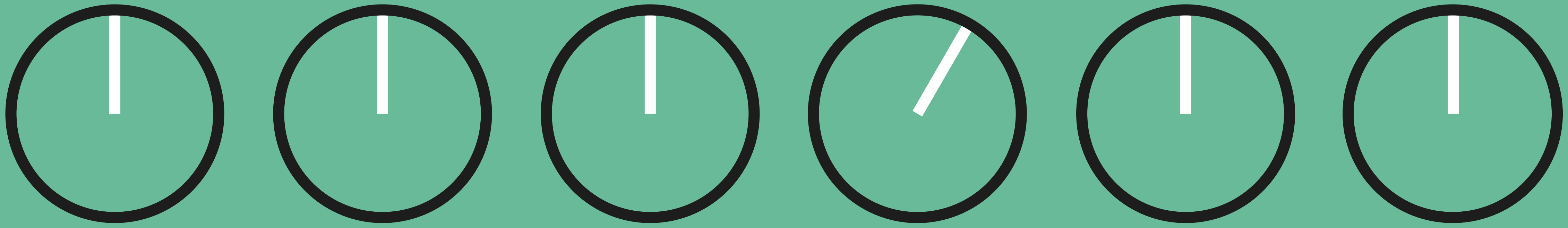
More preattentive visual features

line orientation
length
width
size
curvature
number

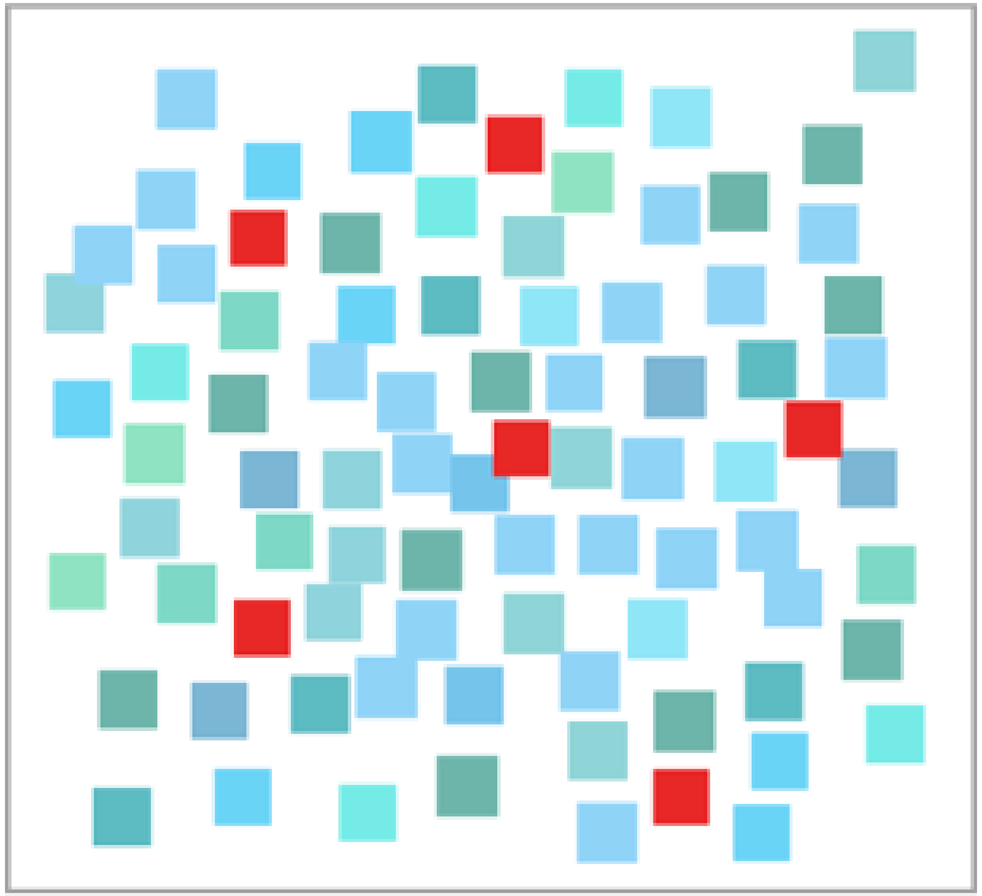
terminators
intersection
closure
color (hue)
color intensity
flicker / blink

direction of motion
stereoscopic depth
3D depth cues
lighting direction

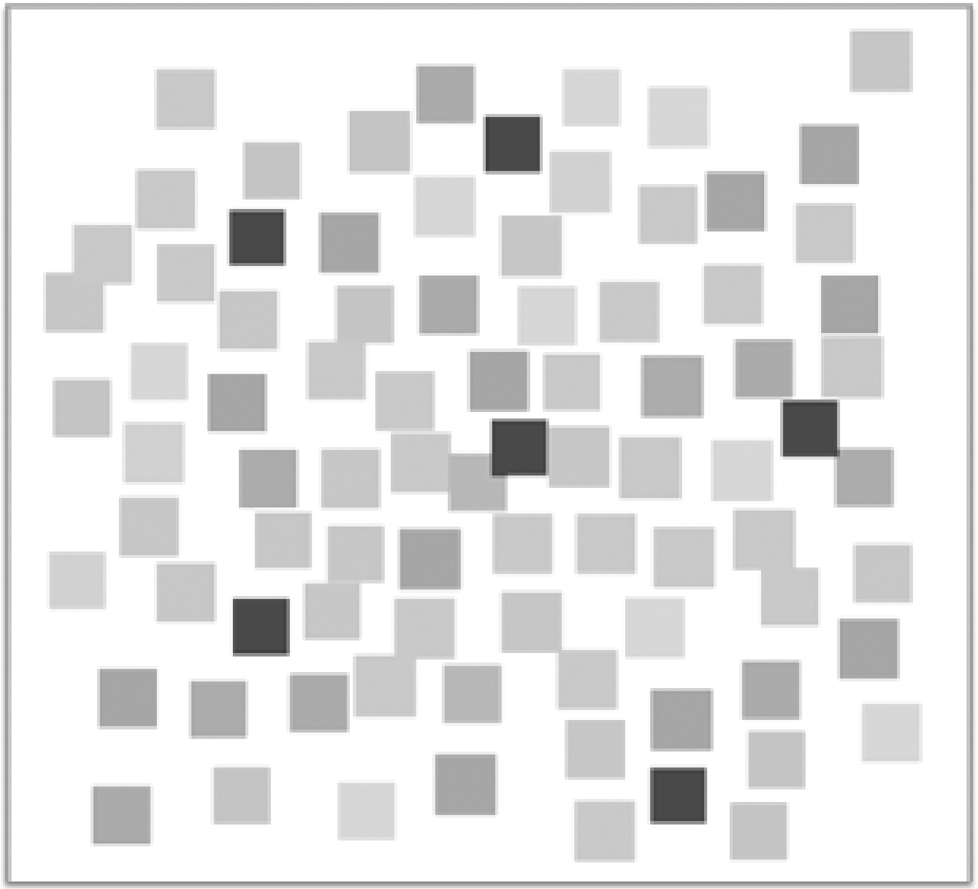
Detection of a **slanted line** in a sea of vertical lines is **preattentive**.



Contrast creates Pop-Out



Hue + Lightness



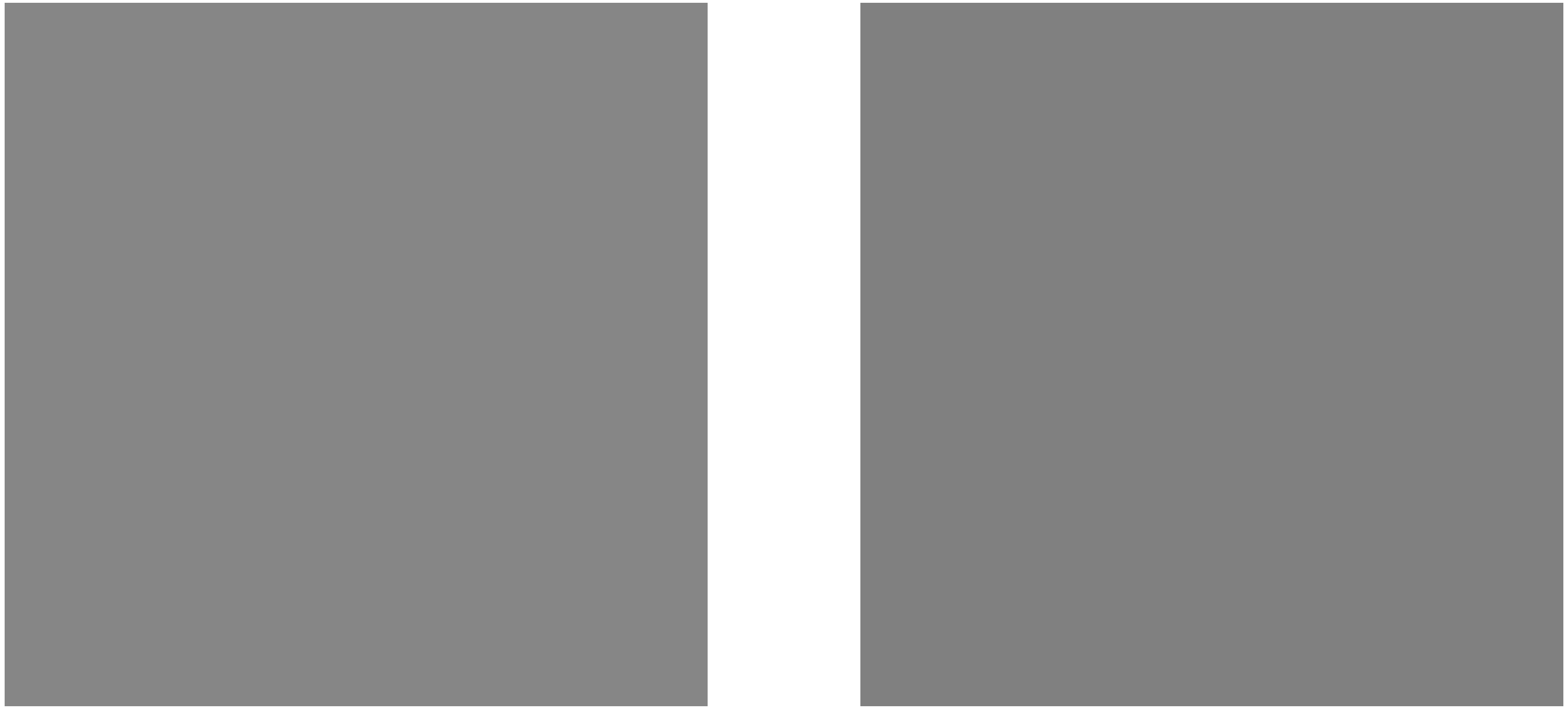
Brightness/contrast only

Thinking with Your Eyes

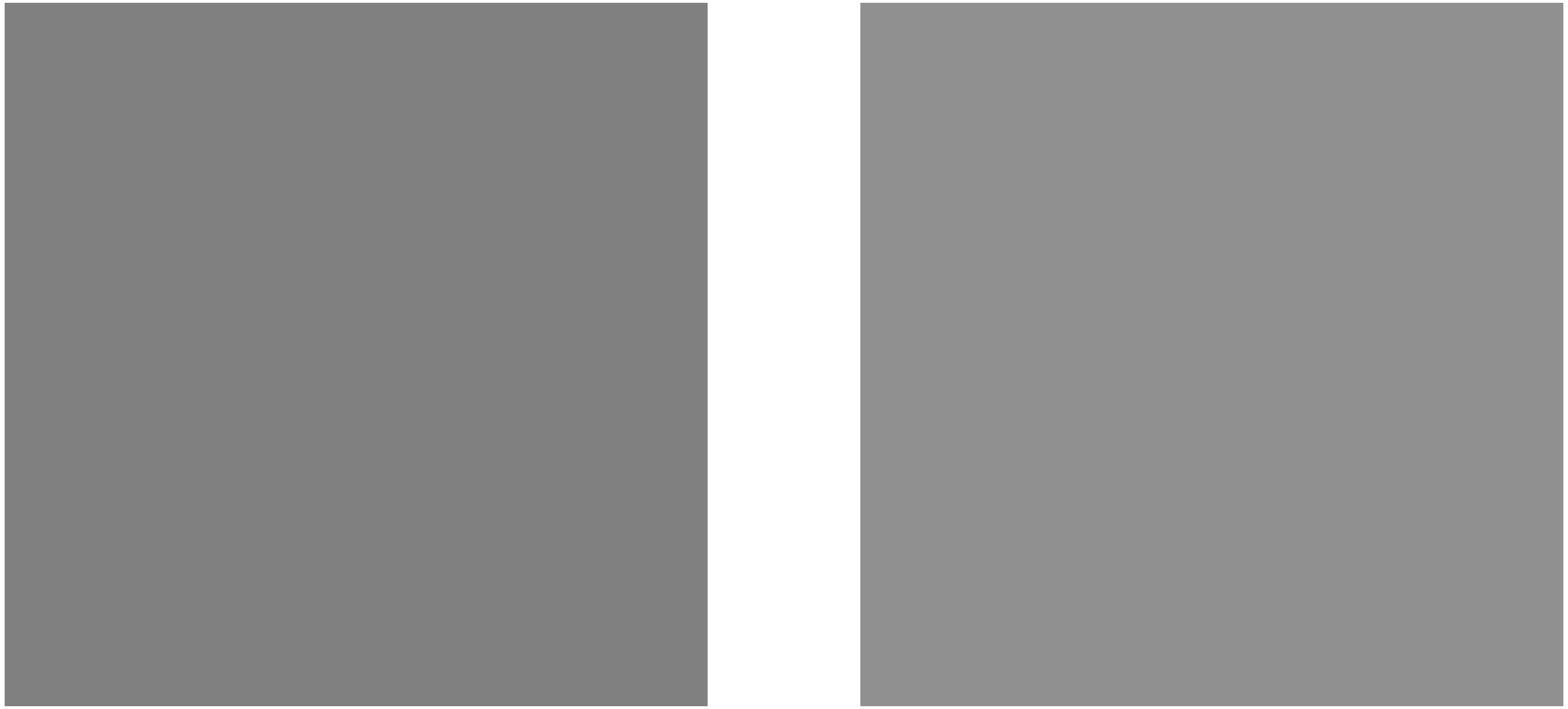
Biology of Vision

Preattentive vs. Attentive

Detection



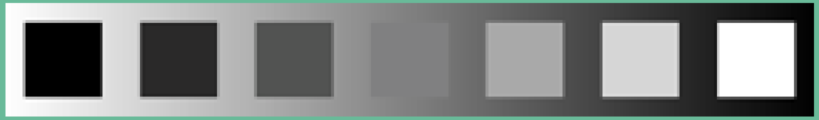
(134, 134, 134) **Δ6** **(128, 128, 128)**



(128, 128, 128) Δ16 (144, 144, 144)

Values are perceived as ordered, so:

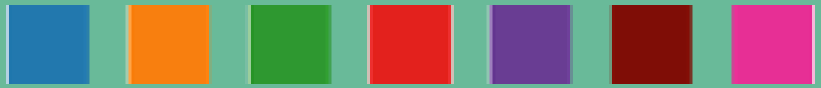
Encode ordinal variables (Ord) by value



Encode continuous variables (Quant) by value

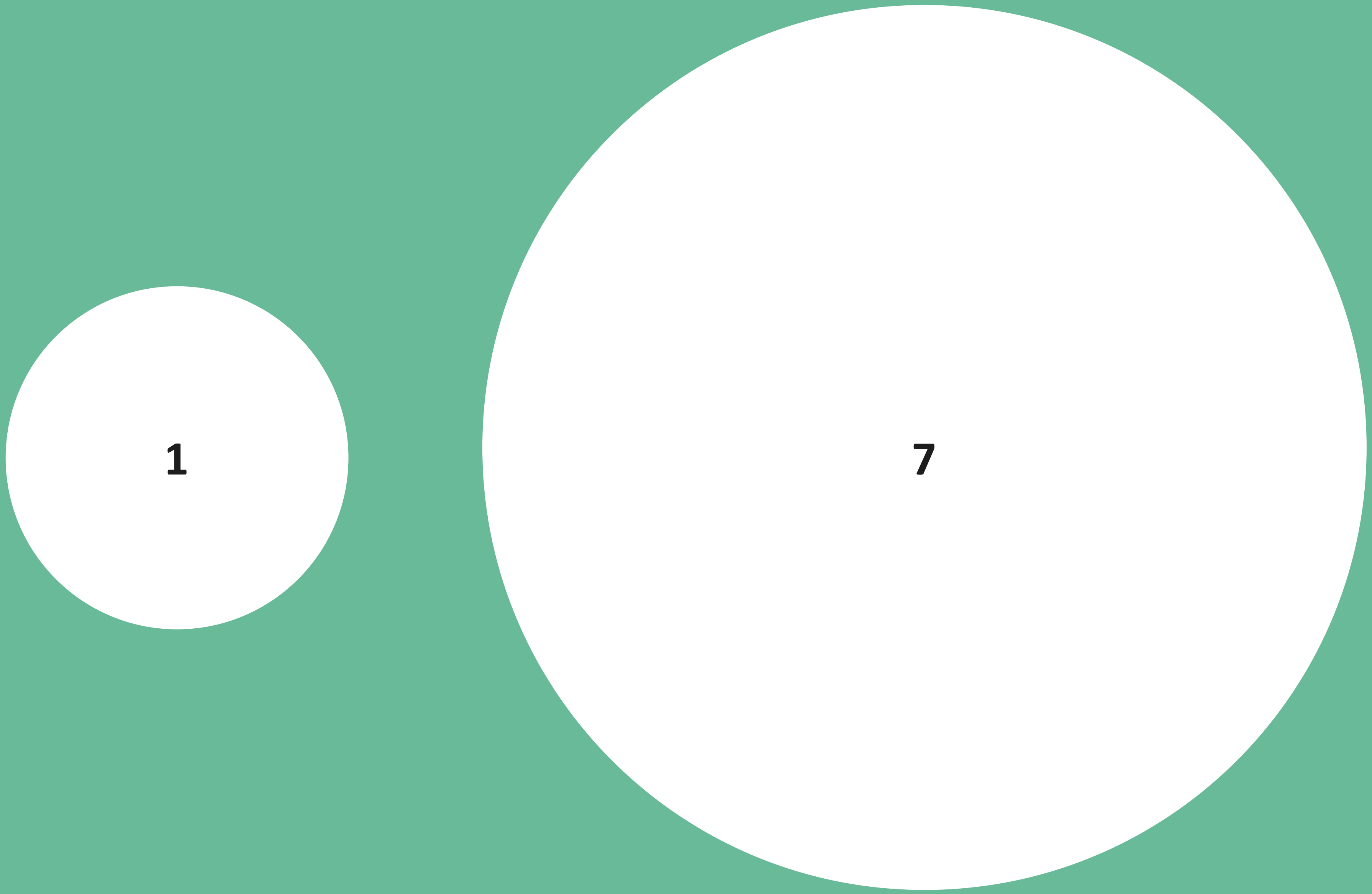
Hue is normally perceived as unordered, so:

Encode nominal variables (N) using color



Estimating Magnitude

Estimating Magnitude



Estimating Magnitude

Most accurate	Position (common) scale
	Position (non-aligned) scale
	Length
	Slope
	Angle
	Area
	Volume
Least accurate	Color (hue/saturation/value)

Integral Dimensions

Two or more attributes of an object are perceived holistically (e.g. width and height of a circular region, as represented by the area).

Difficulty in ignoring one dimension while attending to the other: interference

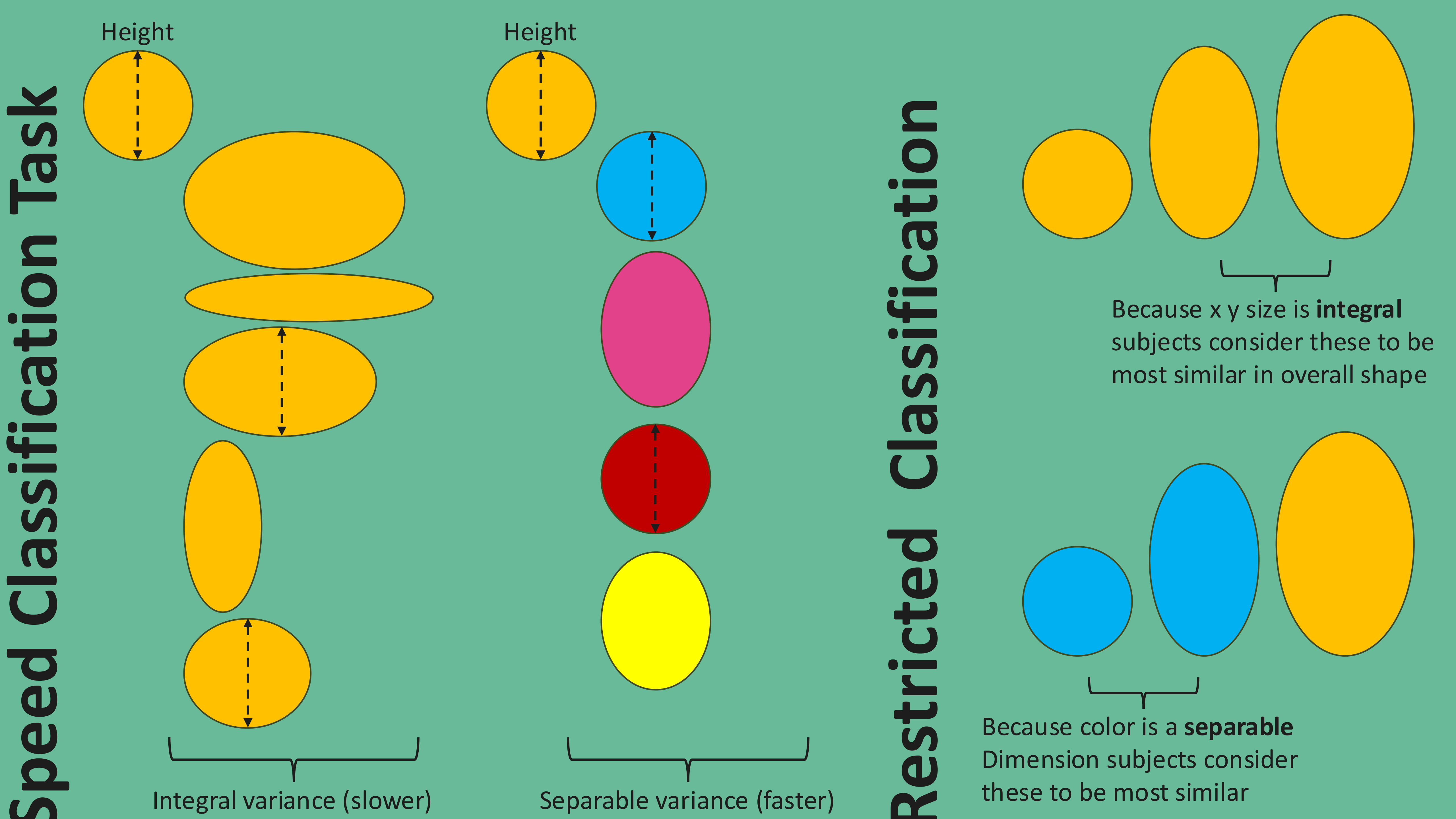
Improvement in evaluating one dimension when the other provides redundant information: redundancy gain

Separable Dimensions

Judged separately, or through analytic processing (e.g. diameter vs. color of a ball, shape vs. color, position vs. color)

Little interference, easy to ignore interference

Little gain with redundant encoding (except for potential accessibility)


Speed Classification Task
Height
Height
Integral variance (slower)
Separable variance (faster)
Restricted Classification
Because x y size is **integral** subjects consider these to be most similar in overall shape
Because color is a **separable** Dimension subjects consider these to be most similar

Length + Width

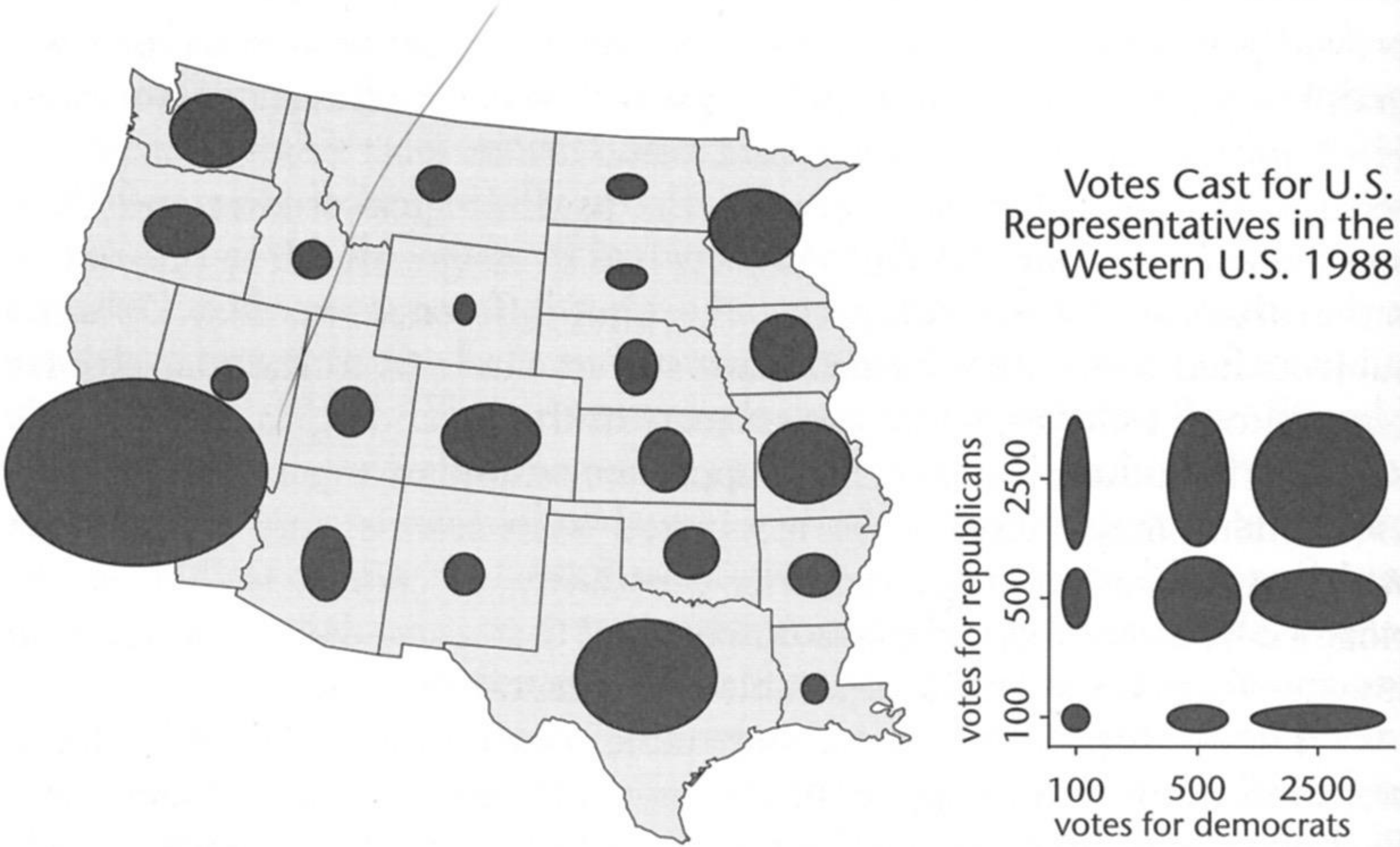


FIGURE 3.38. An example of the use of an ellipse as a map symbol in which the horizontal and vertical axes represent different (but presumably related) variables.

Orientation + Size

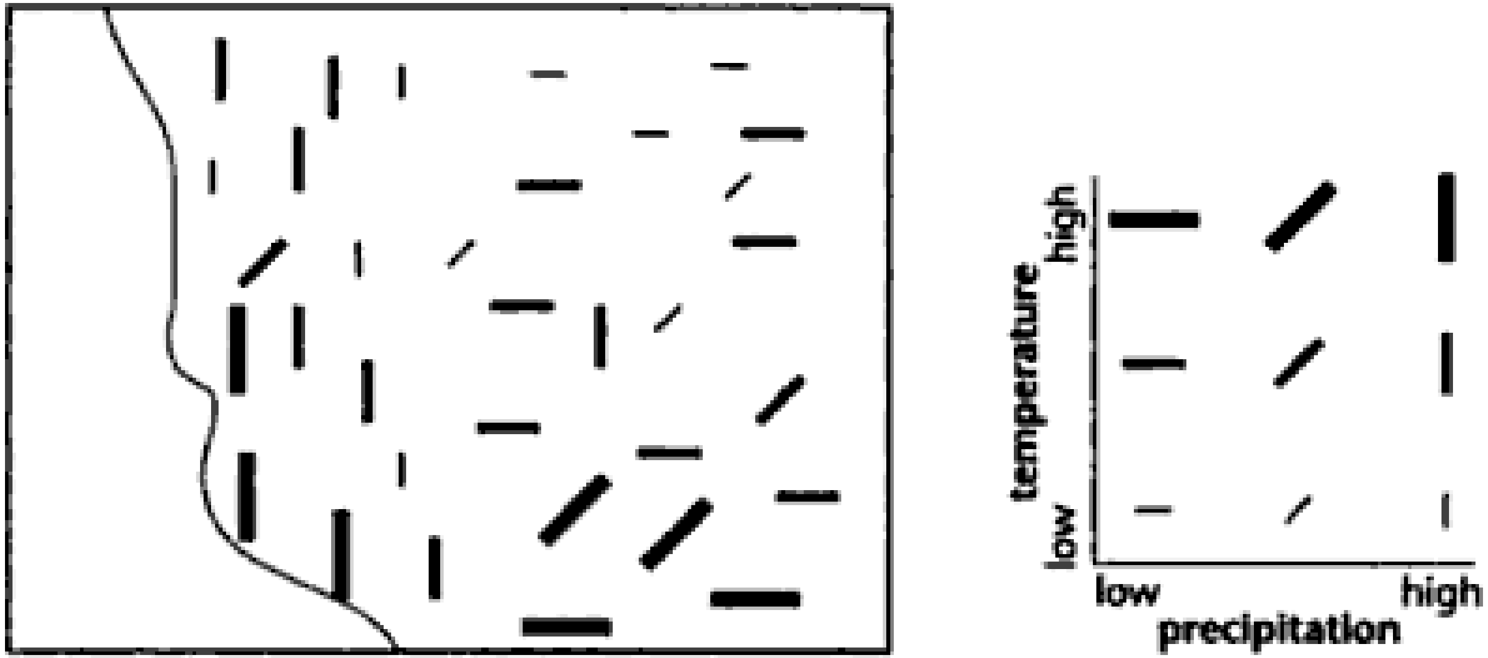


FIGURE 3.36. A map of temperature and precipitation using symbol size and orientation to represent data values on the two variables.

Shape + Size

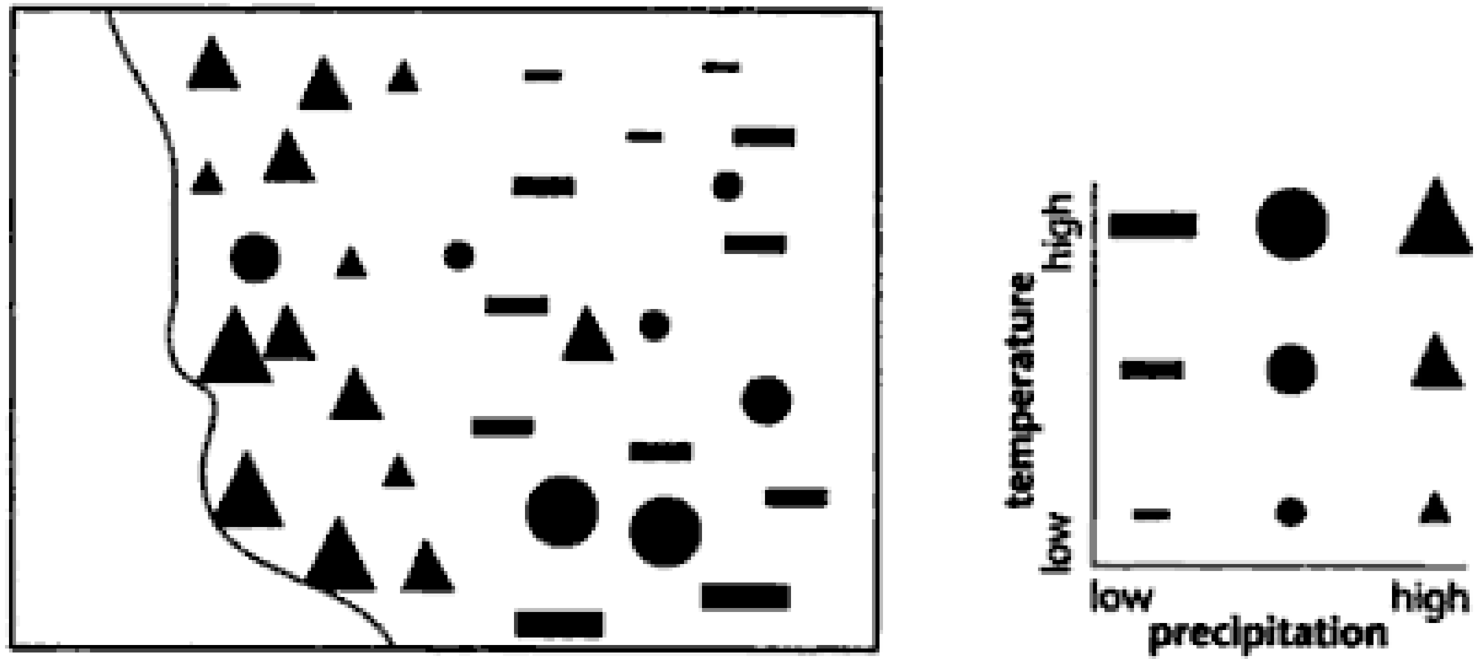
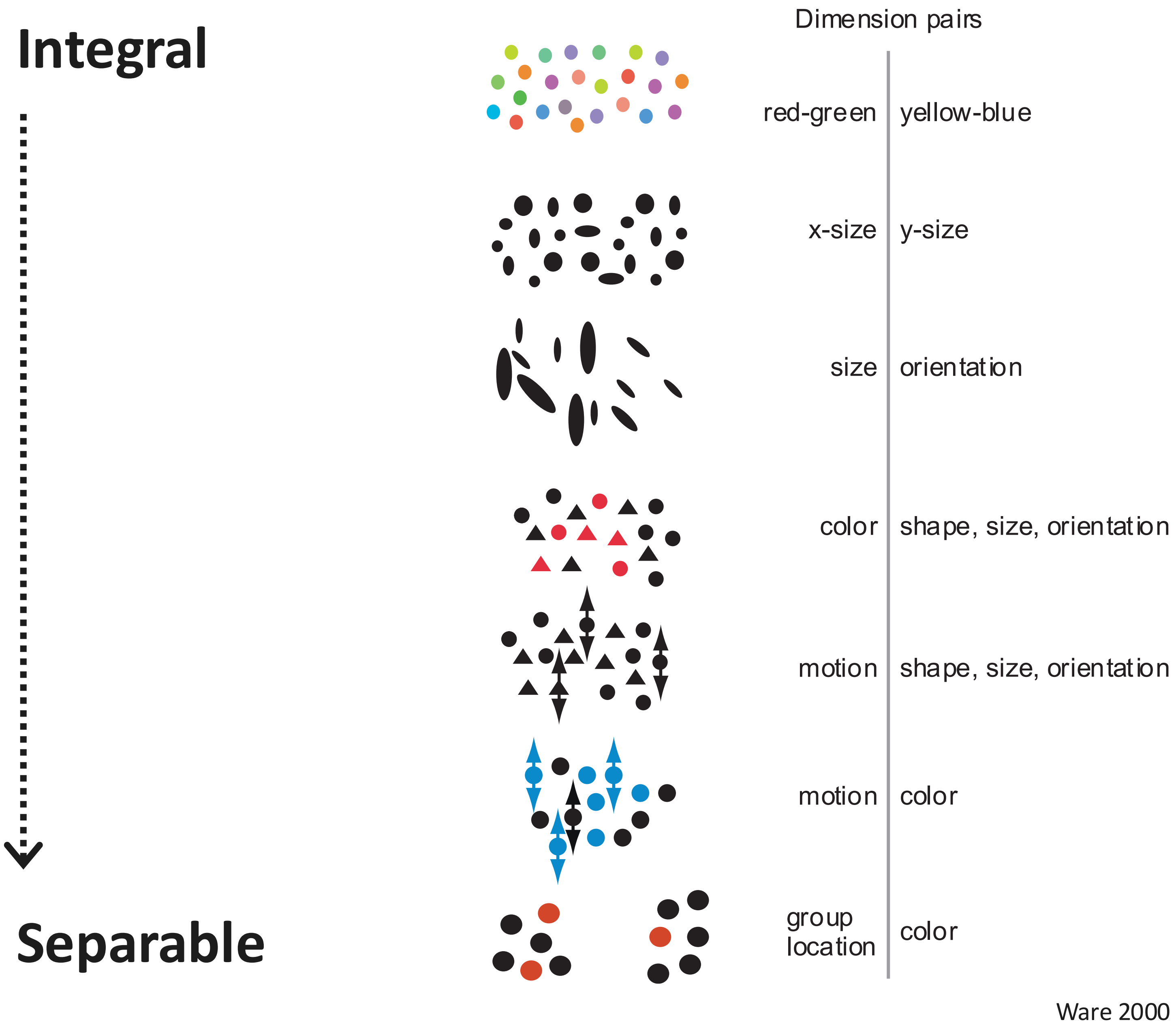


FIGURE 3.40. The bivariate temperature–precipitation map of Figure 3.36, this time using point symbols that vary in shape and size to represent the two quantities.


Integral
Separable
Dimension pairs
red-green | yellow-blue
x-size | y-size
size | orientation
color | shape, size, orientation
motion | shape, size, orientation
motion | color
group location | color
Ware 2000

Break

until x:xx

Group Time

Ideas:

Proposal Revision

Feedback from Doug & Donghoon

A3: Exploratory Data Analysis (due week 6)